

Ex.No. 15: AI-based Traffic Prediction for 5G Networks

Objective 1 Matlab Code and Output:

```
clc;

clear;

close all;

% Time samples

t = 1:10;

% Prediction Accuracy (%)

Accuracy = [90 91 92 93 94 95 96 97 98 99];

% Actual Network Traffic (Mbps)

Traffic = [45 50 55 60 65 70 75 80 85 90];

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(t,Accuracy,'-o','LineWidth',2)

title('AI Prediction Accuracy over Time')

xlabel('Time Interval')

ylabel('Prediction Accuracy (%)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

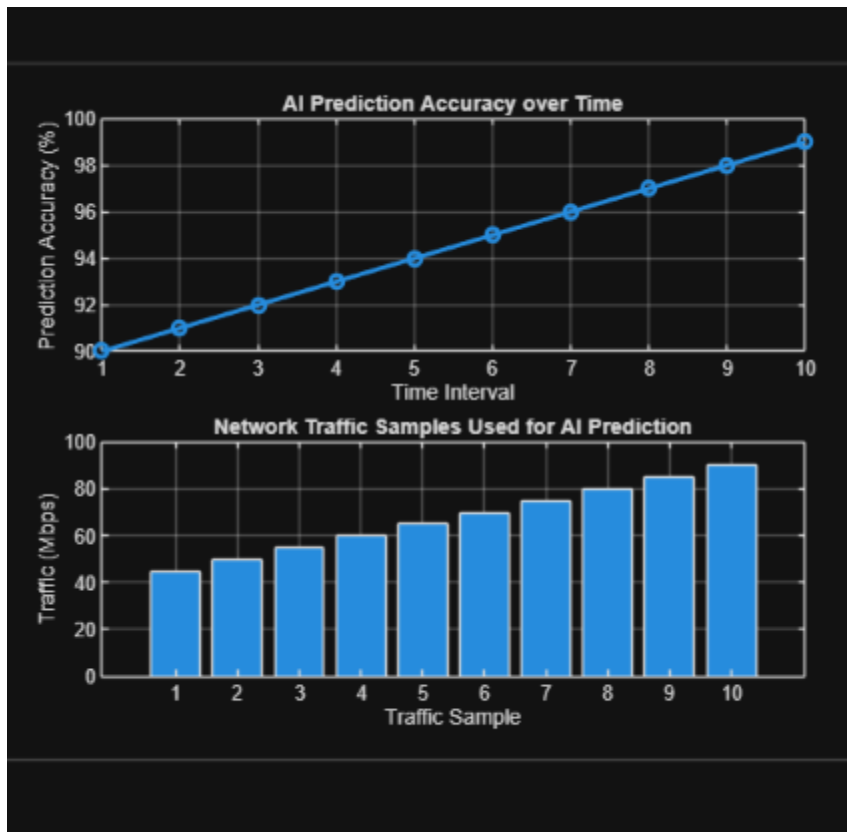
bar(Traffic)

title('Network Traffic Samples Used for AI Prediction')

xlabel('Traffic Sample')

ylabel('Traffic (Mbps)')

grid on
```



Objective 2 Matlab Code and Output:

```
clc;

clear;

close all;

% Time intervals

t = 1:10;

% Actual and Predicted Traffic (Mbps)

Actual = [50 55 60 62 68 70 75 80 85 90];

Predicted = [49 54 61 63 67 71 74 81 84 89];

% Squared Error

SqError = (Actual - Predicted).^2;

% Mean Squared Error

MSE = mean(SqError);

disp(['Mean Squared Error (MSE) = ', num2str(MSE)])
```

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(t,SqError,'-o','LineWidth',2)

title('Squared Error of AI Traffic Prediction')

xlabel('Time Interval')

ylabel('Squared Error')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

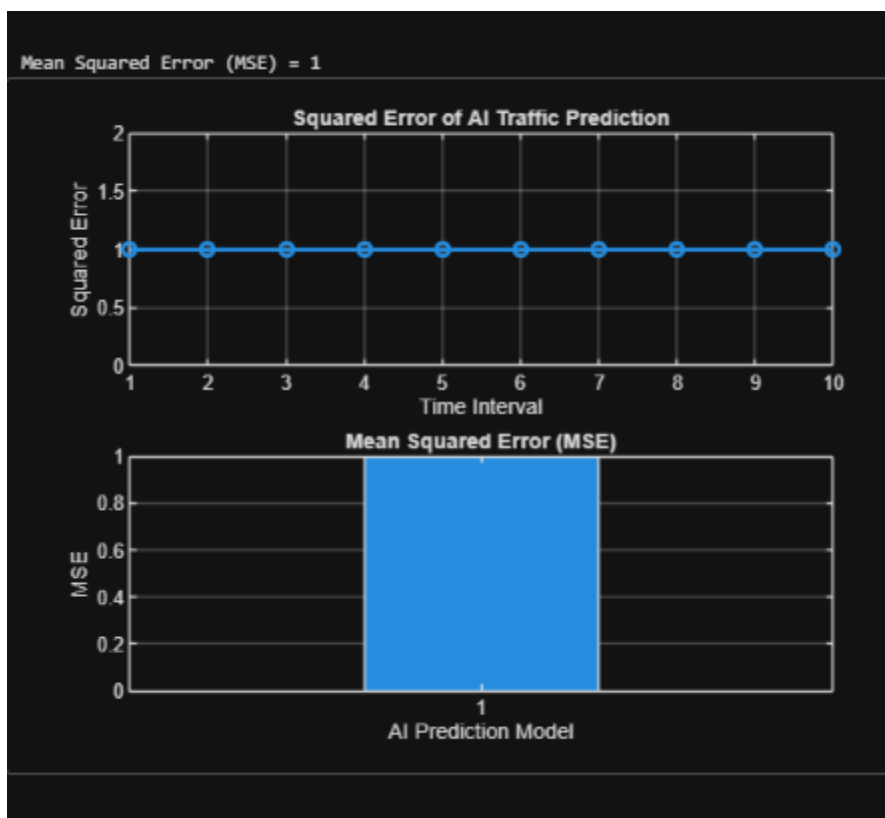
bar([MSE])

title('Mean Squared Error (MSE)')

xlabel('AI Prediction Model')

ylabel('MSE')

grid on



Objective 3 Matlab Code and Output:

```
clc;

clear;

close all;

% Time intervals

t = 1:10;

% Network Throughput (Mbps)

Throughput = [120 135 150 165 180 195 210 225 240 255];

% Network Utilization (%)

Utilization = [60 65 70 74 78 82 86 90 94 98];

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(t,Throughput,'-o','LineWidth',2)

title('AI-Based Network Throughput')

xlabel('Time Interval')

ylabel('Throughput (Mbps)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

plot(Throughput,Utilization,'-s','LineWidth',2)

title('Throughput vs Network Utilization')

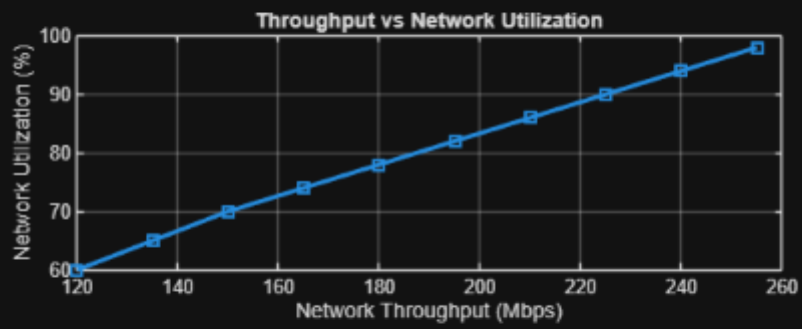
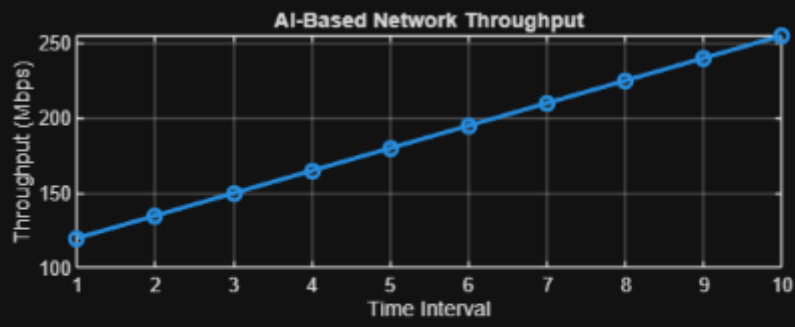
xlabel('Network Throughput (Mbps)')

ylabel('Network Utilization (%)')

grid on

disp('Time Throughput(Mbps) Utilization(%)')

disp([t Throughput Utilization])
```



Time	Throughput(Mbps)	Utilization(%)
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1	120	60
2	135	65
3	150	70
4	165	74
5	180	78
6	195	82
7	210	86
8	225	90
9	240	94
10	255	98